# What Motion Can You Detect?

**SYNOPSIS**

Before we can talk about forces, we have to settle something surprisingly slippery: which motions are real, and which only look real because of where you happen to be standing. The answer turns out to decide the shape of every law in this module — including why Newton's second law is about *acceleration* and not about speed.

**BEFORE YOU READ ON — PREDICT**

You are a passenger in a windowless van with no instruments, riding on a perfectly smooth, straight road. Two questions:

Can you tell, from inside, whether the van is **parked** or **cruising at 60 mph**? And can you tell whether the driver has just **hit the brakes**?

Commit to yes or no for each, and one sentence on how you would find out, before you read on.

**PERFORMANCE EXPECTATIONS**

1. **Decide**, for a described situation inside a closed vehicle, whether the motion could be detected from inside it — and say what experiment would settle it.
2. **State** the principle of relativity in your own words, and explain how it follows from inertia.
3. **Explain** why static and dynamic equilibrium are the same physical state, not two states.
4. **Explain** why Newton's second law relates force to acceleration rather than to velocity.
5. **Draw** a free-body diagram for a single body, and say what belongs on it and what does not.

## Three experiments you have already run

### 1 · Drop a ball in the van

Hold a ball at shoulder height and let go. Where does it land — at your feet, or somewhere behind you, since the van moved forward while the ball was falling?

**At your feet. Exactly as it would if the van were parked in a driveway.**

### 2 · Coffee at 900 kilometres an hour

A cup of coffee on a tray table at cruising altitude sits perfectly still. The aircraft is moving faster than almost anything you will ever ride in, and the surface of the coffee does not know it. Then the pilot throttles back, or hits turbulence, or lands — and it slops over the rim.

**Speed: undetectable. Change of speed: unmistakable.**

### 3 · Standing on the subway

Watch the standing passengers. They lean back as the train pulls out, and forward as it brakes into the next station. In between — at forty miles an hour through a tunnel — they stand up straight and read their phones.

**Nobody braces against speed. Everybody braces against a change in it.**

## What the three have in common

1. State the **Galilean principle of relativity** in your own words.

2. How is the principle of relativity connected to the idea of **inertia**?

3. State **Newton's first law**, and describe why it contains both of the ideas above.

**THE CLAIM THIS NOTE RESTS ON**

There is no experiment you can perform to decide which frame is *really* at rest. **Velocity cannot be measured absolutely — only relative to something else. Acceleration can.**

Which is why Newton's second law is written $F = ma$ and not $F = mv$. A law built on velocity would give different answers to different observers, and would not be a law at all. A law built on acceleration gives the same answer to everyone — so acceleration is the quantity a force can be held responsible for.

**WITH CALCULUS**

Everything above is an argument from examples. It is also a derivation, and the derivation is two lines long.

Let frame $S'$ move along $x$ at a constant velocity $V$ relative to frame $S$. The two observers put the same event at

$$x' = x - Vt$$

1. Differentiate once with respect to $t$. Write $v'$ in terms of $v$.

2. Differentiate again. Write $a'$.

3. Which of those two steps made $V$ disappear — and what property of $V$ was responsible?

4. Now let the second frame **accelerate**, so that $x' = x - \frac{1}{2}At^2$. What is $a'$ this time? What does your answer say about which frames the principle of relativity actually covers?

## Does it hold? Four consequences to check

1. What is a **static equilibrium**? Give an example.

2. Can many forces act on an object that is in static equilibrium? Explain with a diagram.

3. "An object moving at constant velocity has zero net force, so the forces are in equilibrium." True or false, and why?

4. What is a **dynamic equilibrium**, how does it differ from a static one — and **what is the same in both**?

THE TOOL FOR THE REST OF THE MODULE

## The free-body diagram

One body. Every force acting *on* it. Nothing else — not the forces it exerts on other things, not the forces acting on its surroundings, not the motion it happens to have.

That restriction sounds fussy now and will save you repeatedly later. It is the reason a force and its Newton's-third-law partner can never appear on the same diagram: they act on two different bodies, so they belong to two different diagrams.

1. What is a **free-body diagram**? Give two examples.

BACK TO YOUR PREDICTION

You predicted whether you could tell, from inside the windowless van, that it was moving — and whether you could tell it had braked. Which of your two answers was right, and which of the three experiments settled it?

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# Phenomena Illustrating the Laws of Motion

**SYNOPSIS**

This note runs backwards from the others. There is no equation to reach and nothing to derive. You watch four things happen, write down what you notice before you know what to call it, and only then ask which of Newton's laws is doing the work. Two of the four you will be able to settle today. **Two of them you will not** — and knowing exactly why you cannot settle them yet is worth as much as settling them.

**BEFORE YOU READ ON — PREDICT**

A penny is sitting on a card, and the card is resting across the mouth of a cup. Using **one finger**, and without touching the penny, get the penny into the cup.

Write down the method you would try, and what you expect the penny to do, before you turn the page.

**PERFORMANCE EXPECTATIONS**

1. **Record** an observation before explaining it — what moved, how it moved, and what changed — without reaching for vocabulary you have not justified.
2. **Name** the object of interest in a demonstration, and **explain** why the choice of object decides which forces appear in the free-body diagram.
3. **Identify** which of Newton's laws accounts for a demonstrated outcome, and **state** what you would see instead if the force in question were absent.
4. **Distinguish** a phenomenon you can account for with the tools you have from one you cannot, and say **which tool is missing**.

# How to work a phenomenon

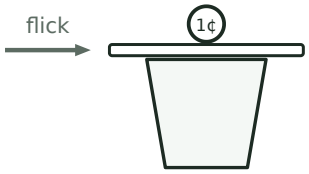
**THE FOUR QUESTIONS**

object → motion → energy → force

**PHENOMENON 1 — A PENNY INTO A CUP**

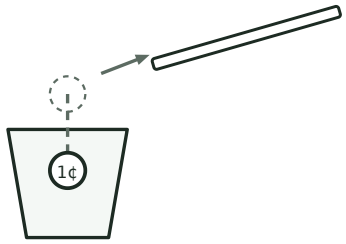
## Flick the card

BEFORE




penny on a card, card on the cup

AFTER




the card leaves sideways. the penny does not.

1. Write at least three things you **notice** and three things you **wonder**. No explanations yet.

| ASK | YOUR ANSWER |
|---|---|
| Object of interest<br>What is making the interesting motion? | |
| The motion<br>Speeding up, slowing down, or no change? Straight or not? | |
| Energy<br>More or less energy in the object? Where did it come from? | |
| Force(s)<br>Which force, on which body, and what is it doing? | |

2. The only thing touching the penny is the card, and the card is only in contact with it for a few thousandths of a second. What force acts on the penny during that time, and roughly how large an effect can it have?

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3. Now answer the question the demonstration asks: **why does the penny not go with the card?** Name the law.

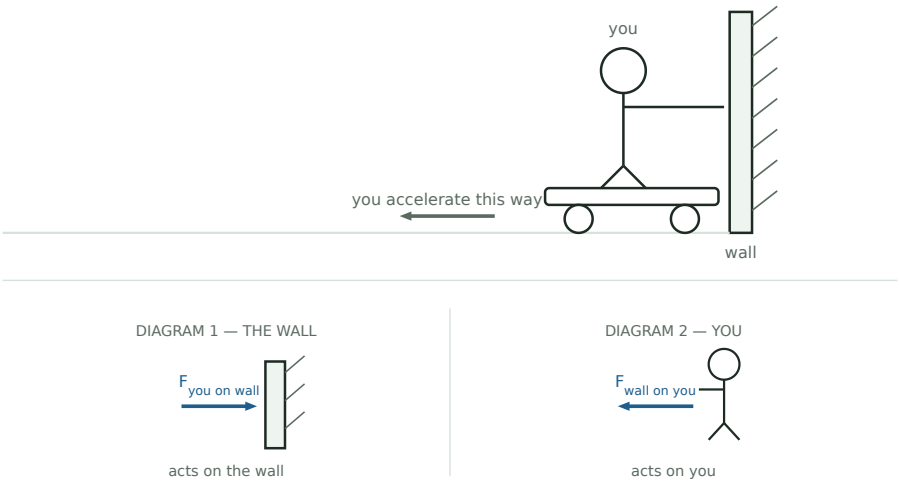
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4. Predict: if the card and the penny were glued together, what would happen? What if the card were made of ice and the penny sat in a puddle of oil?

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PHENOMENON 2 — PUSHING OFF ON A TROLLEY

# Why can you not move without pushing off something?




5. Three things you notice, three things you wonder.

| ASK | YOUR ANSWER |
|---|---|
| Object of interest<br>What is making the interesting motion? | |
| The motion<br>Speeding up, slowing down, or no change? Straight or not? | |
| Energy<br>More or less energy in the object? Where did it come from? | |
| Force(s)<br>Which force, on which body, and what is it doing? | |

6. You are the object of interest. **Which force in the picture accelerates you?** Be precise about which body it acts on.

---

7. You pushed the wall and the wall pushed back with the same size force. The wall did not move. Explain that without saying the forces were unequal.

---

8. Now change the object of interest to **the wall**. Which forces appear in its diagram? Does your answer to the previous question change?

---

9. Push harder. What happens to your acceleration, and what does that tell you about the form of the second law?

---

10. Predict: on a perfectly frictionless floor with nothing within reach, could you get yourself moving at all? What if you were holding a heavy backpack?

---

PHENOMENA 3 AND 4 — LEFT OPEN ON PURPOSE

## Two you cannot finish yet

### OPEN PHENOMENON 3 — BOUNCING A BALL ON A BALL

**Why does the small ball bounce so much higher when it is dropped on top of a large bouncy ball than when it is dropped on its own?**

- Three things you notice:
- Your best guess, in one sentence:

**Why not now.** Answering this properly means tracking what each ball carries into the collision and how that is shared out in the instant they touch. Forces alone will not get you there — you need momentum and the impulse of a very short contact, which is a later module. We will come back to this demonstration by name when we have them.

### OPEN PHENOMENON 4 — SHOOTING A FALLING TARGET

**A target is released at the exact instant the ball is fired straight at it. Why do you hit it every time, no matter how fast you fire or how far away the target is?**

- Three things you notice:

- Your best guess, in one sentence:

**Why not now.** This one is closer than it looks. You already know from Module 2 that the horizontal and vertical parts of a projectile's motion do not interfere with each other. The missing piece is the claim **N4** is about: every object near the Earth falls with the same acceleration, whatever its mass. Put those two together and the result stops being a trick. Hold this one — we settle it in N4.

### BACK TO YOUR PREDICTION

You predicted a method for getting the penny into the cup. Did it work? Name the property of the penny that made it work, and say what would have happened had you moved the card slowly.

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# Putting Numbers on the Phenomena

**SYNOPSIS**

In **N2** you watched things happen and said which law was responsible. That is half of physics. The other half is *how much* — and it is the half that decides whether a bridge stands up. This note goes back to the same demonstrations and works out the numbers: what the penny weighs, how hard each person on a trolley accelerates, and where to point the launcher. The physics is not new. What is new is the discipline of getting an answer you can defend.

**BEFORE YOU READ ON — PREDICT**

You are on a trolley with a total mass of **100 kg**. You push on someone else whose trolley and body come to **200 kg**. Two separate questions, and they do not have the same answer:

Do the two of you feel the **same size force**?

Do the two of you get the **same acceleration**?

Answer both, yes or no, with one sentence each, before you read on.

**PERFORMANCE EXPECTATIONS**

1. **Convert** a mass given in grams to kilograms before substituting it, and **show** the conversion as a step rather than doing it in your head.
2. **Carry** units through every line of a calculation and **state** the unit of the answer. An answer without a unit is not an answer.
3. **Compute** the weight of a mass from w = mg.
4. **Apply** a = F/m to each member of an interacting pair, and **explain** why the two accelerations differ although the two forces do not.
5. **Use** trigonometry to find a launch angle from a height and a distance.
6. **Check** whether an answer is reasonable before writing it down, and say what made you accept it.

MOVEMENT 1 — THE RULES OF THE ROAD

## How to get an answer you can defend

**EVERY TIME**

convert → substitute → carry the units → check it is reasonable

MOVEMENT 2 — WHAT THE PENNY WEIGHS

## From mass to weight

**Eq. 1** $w = F_g = mg$

g = 9.8 m/s² near the Earth's surface

**TASK 1 — A PENNY**

1. The mass of a penny is **2.5 g**. Convert it to kilograms, and write the conversion as its own line.

2. Find its weight. Show the substitution with units attached.

3. Is the answer reasonable? Say what you compared it with.

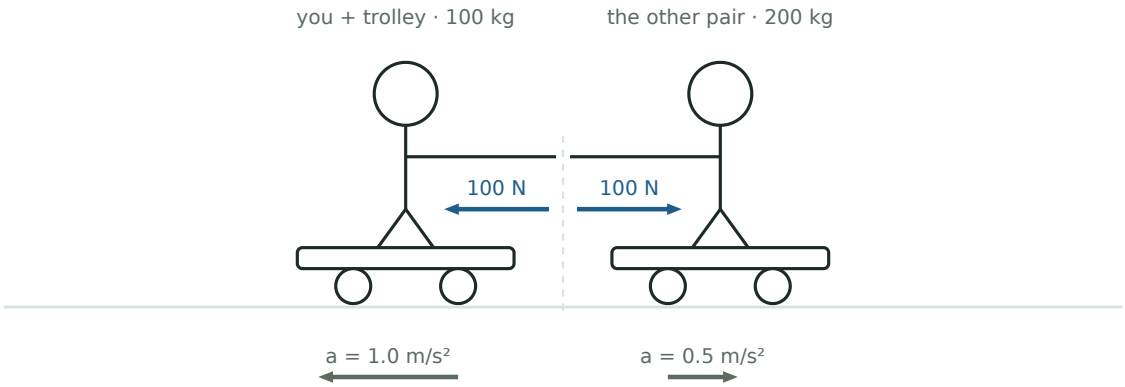
**TASK 2 — AN ADULT**

1. A typical adult has a mass of **70 kg**. Find the weight, with units on every number.

2. Your answer is a number of newtons. Roughly how many pounds is that, and does the comparison make sense to you?

MOVEMENT 3 — THE THIRD LAW, WITH NUMBERS

## Equal forces, unequal accelerations




the force arrows match. the acceleration arrows do not.

TASK 3 — BOTH SIDES OF THE PUSH

1. You and your trolley come to 100 kg. You push with 100 N. Find your acceleration.

2. The other person and their trolley come to 200 kg. What force acts on *them*? State it before you calculate anything.

3. Find their acceleration.

4. Compare the two accelerations. Write the comparison as a ratio, and say what that ratio is the ratio *of*.

5. Go back to the wall in N2 and finish the argument properly: name the quantity that makes the wall's response invisible, and say what would have to be true for you to see the wall move.

6. In **N1** you met the claim that velocity is relative but acceleration is absolute. You and the other person disagree about who is moving. Do you disagree about the *accelerations*? Use your two numbers to answer.

**WITH CALCULUS**

$\Sigma F = ma$ is not the most general form of the second law. Newton wrote it about **momentum**, $p = mv$:

$$\Sigma F = \mathrm{d}p / \mathrm{d}t$$

Which turns into the equation you have been using all note, but only after an assumption that nothing on this page has stated out loud.

1. Substitute $p = mv$ and differentiate the **product**. You should get two terms.

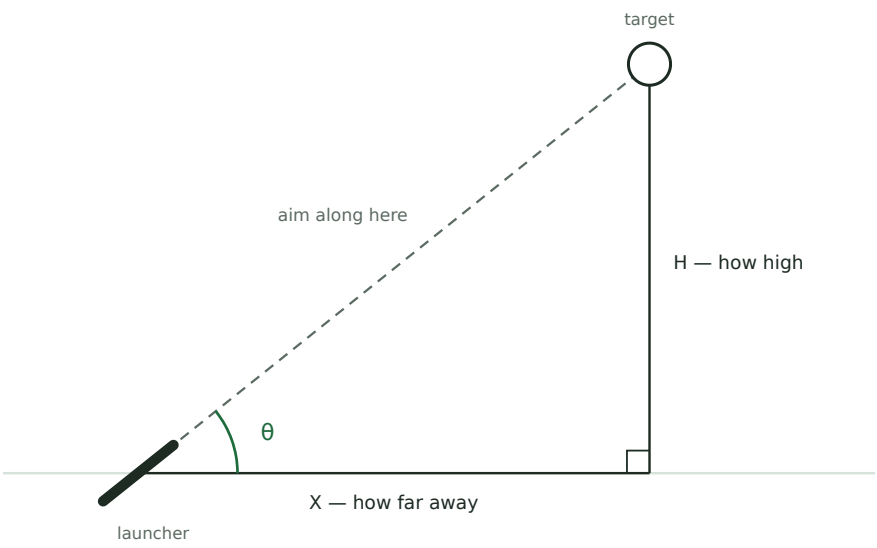
2. One of them is $ma$. What is the other, and under what condition is it zero?

3. That condition is the assumption. It held in every calculation in this note — say why, and then name a situation in which it would **fail**.

4. Back to **N1**. You showed there that two inertial observers agree about $a$. Do they agree about $p$? Do they agree about $\mathrm{d}p/\mathrm{d}t$? Which of those three answers is the reason $\Sigma F = \mathrm{d}p/\mathrm{d}t$ can be a law at all?

MOVEMENT 4 — AIMING THE LAUNCHER

# Where to point the barrel




## TASK 4 — THE LAUNCH ANGLE

1. Draw the setup in your own notes and label X, H and θ.

2. Which pair of sides does θ sit between, and therefore which trigonometric ratio do you need? Write the ratio before putting any numbers in.

3. With X = 2.5 m and H = 2.0 m, find θ.

4. Check it: is your angle less than or greater than 45°, and does that match the picture?

## BACK TO YOUR PREDICTION

You answered two questions at the top: whether the forces are equal, and whether the accelerations are equal. Which of your two answers was right? If you gave the same answer to both, name the step in Task 3 that separates them.

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# Weight Is a Force of Gravity on a Mass

**SYNOPSIS**

In everyday speech “weight” is a number on a bathroom scale, reported in pounds — a unit of *mass*. In physics it is something else entirely: weight is a **force**, measured in newtons, the pull of a planet on a mass. Mass belongs to the object and travels with it. Weight does not: it is what a *particular planet* does to that object. This note measures the relationship between the two, finds that it is a straight line, and shows why the slope of that line is the reason a feather and a bowling ball fall together in a vacuum.

**BEFORE YOU READ ON — PREDICT**

You hold a **20 kg** mass in one hand and a **10 kg** mass in the other, and release them at the same instant. Gravity pulls *twice as hard* on the 20 kg mass — that much is not in dispute.
Does it reach the floor first?

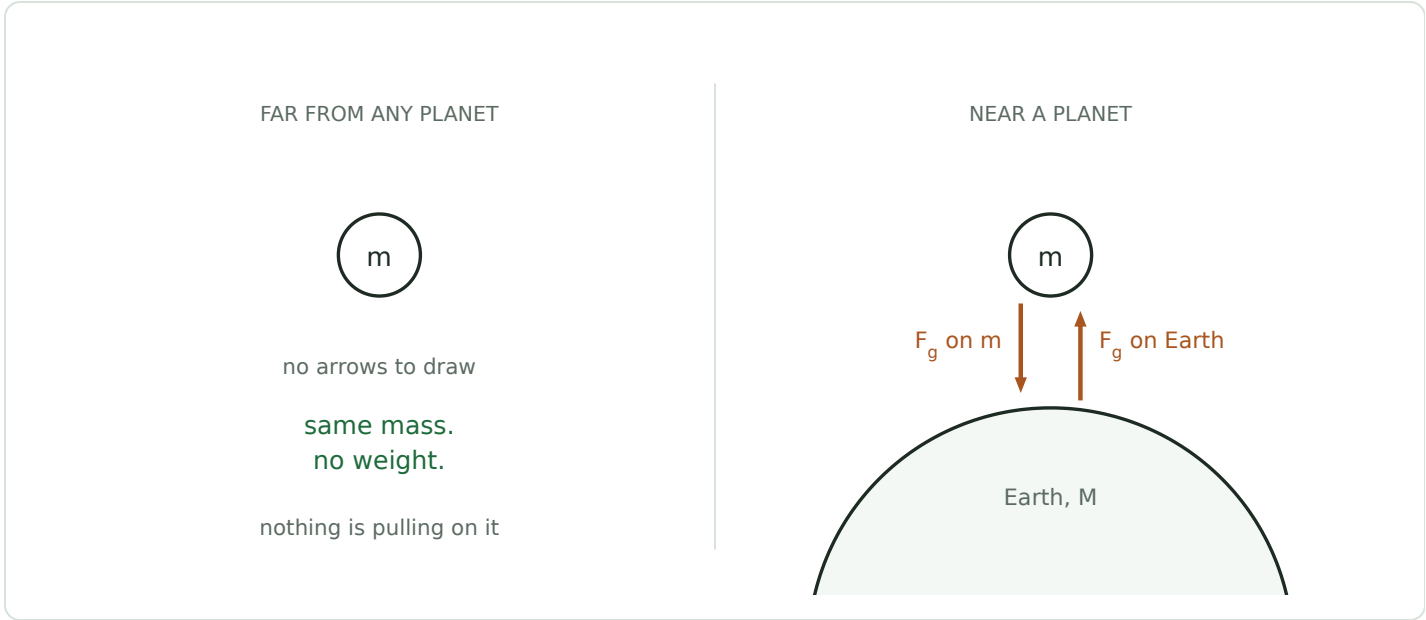
Commit to yes or no, and write one sentence saying why, before you read further.

**PERFORMANCE EXPECTATIONS**

1. **Distinguish** mass from weight by unit, by definition, and by what happens to each when the object is moved to another planet.
2. **Measure** weight against mass with a spring scale, plot the two, and **interpret** the slope of the line, including its unit.
3. **Explain** why every mass near a planet falls with the same acceleration, in terms of force *and* inertia.
4. **Identify** the third-law partner of an object’s weight, and state which body it acts on.
5. **Calculate** weight, mass, normal force, or $g$ given the other quantities, on Earth or on any other planet.

MOVEMENT 1 — WEIGHT IS NOT A PROPERTY OF THE OBJECT

## Where does weight come from?




1. A book rests on a table. Force given: **the table pushes up on the book**.

   partner _____ acts on _____

2. The same book on the same table. Force given: **the Earth pulls down on the book**.

   partner _____ acts on _____

3. The book is now in free fall, halfway to the floor. Force given: **the Earth pulls down on the book**.

   partner _____ acts on _____

   Has your answer changed from the previous question? Should it have?

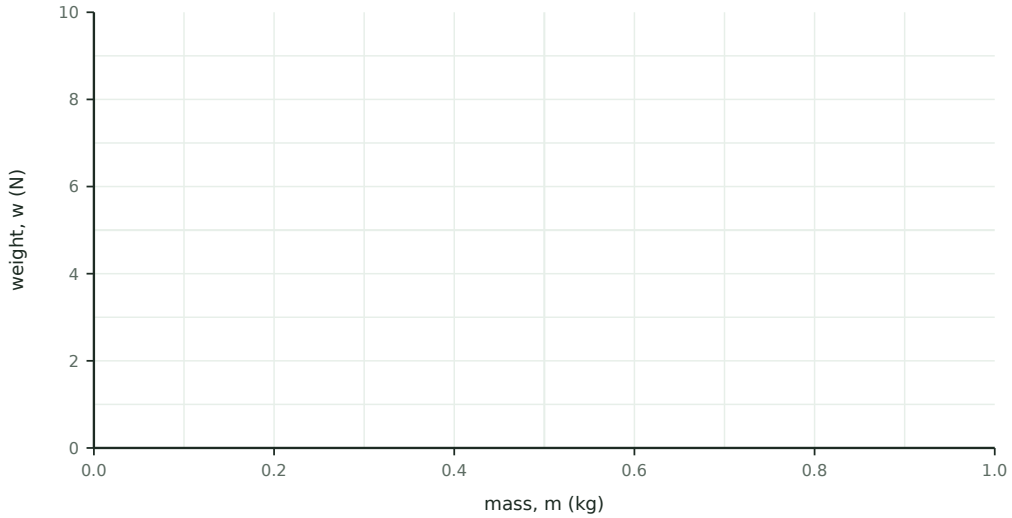
MOVEMENT 2 — MEASURE IT

## Weight against mass

Hang a known mass on a spring scale and read the force it registers. Repeat with different masses. You are asking the scale a simple question: when the mass goes up, what does the pull do?

1. Zero the spring scale with nothing hanging from it.
2. Hang each mass in turn and let it come to rest before you read the needle. Record the reading in **newtons**.
3. Fill in the table. Keep the readings to the precision the scale actually gives you.

| MASS, M (KG) | WEIGHT, W (N) |
|---|---|
| 0.1 | |
| 0.2 | |
| 0.5 | |
| 0.7 | |
| 0.9 | |




Plot weight against mass. Draw the best straight line through your points — and think about whether that line is entitled to pass through the origin.

1. Is the plot linear? What does that tell you about how weight depends on mass?

2. Find the slope, and give its unit. Do not skip the unit — it is the whole point of the next question.

3. What does the slope represent? Where have you seen that number before?

**THE RESULT**

**$w = F_g = mg$**

**MOVEMENT 3 — WHY EVERYTHING FALLS TOGETHER**

## Your prediction, settled

**Eq. 1** $F_g = ma$ — the only force on a freely falling mass is its weight

**Eq. 2** $a = \frac{F_g}{m} = \frac{mg}{m} = g$ — the mass cancels — the acceleration does not depend on it

4. “You drop a 20 kg and a 10 kg mass. The 20 kg mass experiences twice as large a force of gravity as the 10 kg mass, but they still accelerate at the same rate, 9.8 m/s².” True or false?

5. Give the reason for your choice in one sentence. Your sentence must mention **both** force and inertia — an answer that mentions only one of them is not an explanation.

6. Go back to your prediction in the box at the top. Was it right? If not, name the step in the reasoning where it went wrong.

7. Compare this with the guess you wrote in N2. Which part did you already have, and which part were you missing?

8. Fire the ball **twice as fast**. Do you still hit the target? What changes about *where* and *when* you hit it?

9. Now break it. Name a change to the demonstration that would make the shot miss — and say which of the two moves above your change destroys.

MOVEMENT 4 — WEIGHT, NORMAL FORCE, AND OTHER PLANETS

## An object at rest on a level floor

10. An obese cat with a mass of 46 kg rests on the floor. Find its weight and the magnitude of the normal force on it, and draw its free-body diagram.

11. A cat weighs 57.3 N. What is its mass? Resting on a floor, what normal force acts on it? If the cat is sent to another planet, what is its mass there?

12. On the Moon, a person of mass 70 kg weighs 112 N. What is the acceleration due to gravity on the Moon?

13. A 65 kg astronaut weighs 638 N on Earth. What is the astronaut's **mass** on Mars, where $g = 3.4\ m/s^2$? What is the astronaut's **weight** there?

**THE HABIT TO BUILD**

Mass is given in kilograms and never changes. Weight is given in newtons and changes with the planet. If a quantity in a problem is in kg it is a mass; if it is in N it is a force. Check the unit before you decide which one you have been handed.

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# Friction Is Directly Proportional to the Normal Force

**SYNOPSIS**

Without friction, you can't walk. To move forward you push *backward* on the ground, and it is the ground's grip — friction — that pushes you forward in return: Newton's third law in action. That grip depends on just two things: the normal force pressing the surfaces together, and the surface roughness, μ. Note carefully which one: the *normal force*, not the weight. On a level floor they happen to be equal, which is why the two are so easily confused. And note the odd geometry — friction and the normal force are directly proportional yet *perpendicular* to each other: the one case where a vertical force sets the size of a horizontal one.

**BEFORE YOU READ ON — PREDICT**

You drag a loaded sled across the bench at a steady speed, pulling it with a spring scale, and the scale reads some value. Now you double the mass riding on the sled and drag it again, the same way.

What does the scale read — the same, twice as much, or somewhere in between? And what happens to the *coefficient* of friction?

Commit to an answer for both before you turn the page. You will measure it yourself.

**PERFORMANCE EXPECTATIONS**

1. **State** the direction of the friction force for a given relative motion, including the case of walking.
2. **Compute** the friction force from μ and the normal force, and the normal force on a level surface.
3. **Explain** what μ does and does not depend on — and why the normal force, not the weight, is what friction follows.
4. **Decide** whether static friction is at its maximum or below it in a described situation.
5. **Measure** a coefficient of kinetic friction as the slope of a friction-vs-normal-force graph.

6. **Use** the static-to-kinetic force trace to say what happens at the instant sliding begins.

## Your tasks in this note

1. **Free-body diagram:** a person walking forward — where does friction point, and why is that the direction of motion?
2. **Free-body diagram:** a falling parachuter — drag is friction's cousin acting through a fluid.
3. **Write the equations** of kinetic and static friction in terms of $\mu_k$, $\mu_s$, and the normal force.
4. **Work the questions** in your guided notes as the video develops each case.

## The key relationships

| | | |
|---|---|---|
| **Eq. 1** | $f_k = \mu_k N$ | kinetic friction — surfaces sliding past each other |
| **Eq. 2** | $f_s \leq \mu_s N$ | static friction — grips as hard as needed, up to its limit |
| **Eq. 3** | $\mu_s > \mu_k$ | breaking the grip is harder than keeping the slide — why wheels lock and joints creak loose |
| **Eq. 4** | $f = \mu m g$ | **level surface only** — the shortcut you get by substituting N = mg |

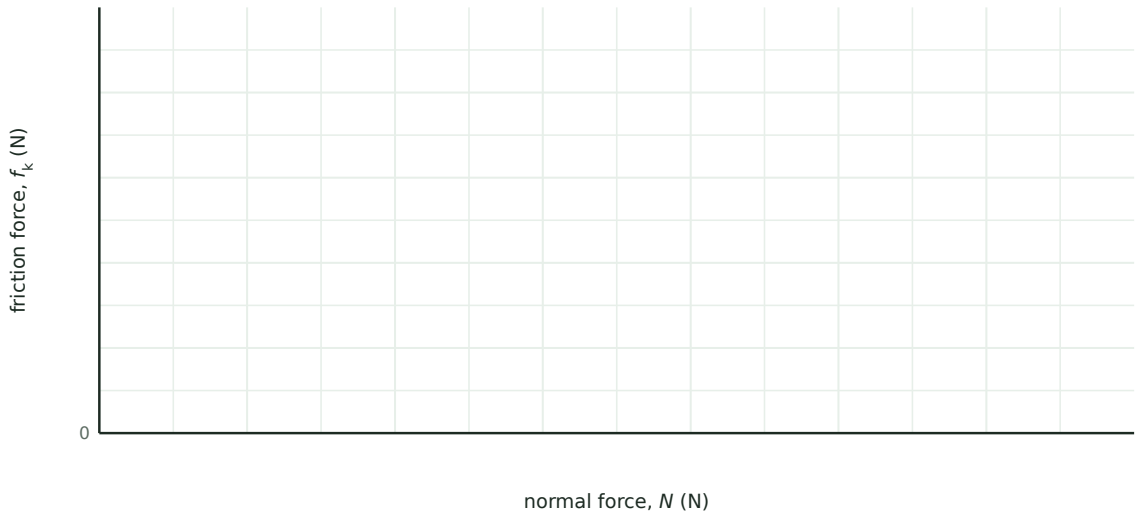
## Measure μ for yourself

A coefficient of friction is not something to look up — it is something you can measure in about ten minutes. You will drag a loaded sled across the bench with a spring scale and find $\mu_k$ as the slope of a graph.

1. **Set up.** Put the sled on the bench with the spring scale hooked to it, the scale held horizontally.
2. **Pull at a steady speed.** This part matters more than anything else in the procedure. Only when the sled moves at *constant velocity* is the acceleration zero, so that the forces balance and the scale reading equals the friction force exactly. If you are speeding it up, the reading is bigger than the friction and your data is wrong. Practise until you can hold it steady.
3. **Watch the scale as it breaks loose.** Pull slowly from rest and note the reading at the instant the sled starts to move — then the lower, steadier reading once it is sliding. Record both.

4. **Add mass and repeat.** Load the sled with five different masses, dragging steadily each time, and record the scale reading for each.

| LOAD | TOTAL MASS, M (KG) | NORMAL FORCE, N = MG (N) | SCALE READING WHILE SLIDING = $F_K$ (N) |
|---|---|---|---|
| sled alone | | | |
| + mass 1 | | | |
| + mass 2 | | | |
| + mass 3 | | | |
| + mass 4 | | | |




Plot the friction force against the normal force. On a level bench the normal force is just the total weight, N = mg — so the third column of your table is the x-axis and the fourth is the y-axis.

1. **Is your plot a straight line through the origin?** If it is, you have just shown that friction is *directly proportional* to the normal force — which is the title of this note, and now it is your measurement rather than my claim.

2. **Find the slope.** What are its units? The slope is $\mu_k$, and the answer to the units question is why $\mu$ is just a number.

3. **Compare your two readings from step 3** — the one at the instant it broke loose, and the steady one while sliding. Which is bigger, and what does that tell you about $\mu_s$ and $\mu_k$?

4. **Now turn the sled on its side**, so it rests on a smaller face, and drag the same load again. Predict first, then measure. Does the reading change?

5. **Now check it.** You doubled the mass. What happened to the scale reading, and what happened to $\mu$?

**BACK TO YOUR PREDICTION**

You predicted what the spring scale would read when you doubled the mass on the sled. Compare that with what your data actually showed, and name the quantity you were really changing.

**WATCH · THE NOTE WORKED THROUGH**

Recording to come.  Print the guided notes (link just below), watch, and fill them in as you go.

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# Verifying Newton's Second Law with a Half Atwood Machine


### SYNOPSIS

Newton's second law fits in five characters, but it makes a testable promise: double the net force on an object and its acceleration doubles; double the mass and the acceleration halves. This note puts that promise on trial with a cart on a track pulled by a hanging mass. The interesting part is that the obvious experiment *does not work* — and finding out why is most of the physics.


### BEFORE YOU READ ON — PREDICT

The cart is pulled by a hanging mass and accelerates. You now **double the hanging mass**, which doubles the force pulling the system along.

Does the acceleration double?

Commit to yes or no, with one sentence of reasoning, before you turn the page.

### PERFORMANCE EXPECTATIONS

1. **Draw** a free-body diagram for each body in a connected two-body system, and mark the positive direction of acceleration on each.
2. **Write** ΣF = ma for each body along its own direction of motion.
3. **Combine** the two equations to predict the system's acceleration, and explain why the total mass appears in the denominator.
4. **Explain** why adding mass to the hanger does not double the acceleration.
5. **Describe** a procedure that tests F = ma by changing the net force while holding the total mass fixed.

### CARRY THIS INTO THE LAB

**This note is Lab 4.** The setup drawing, the two free-body diagrams and the derivation below are not background for that lab — they are its Introduction, its Experimental section and its analysis plan. You are expected to bring them into the report and build on them, section by section:

**Introduction** your own derivation of $a = F_{g,H}/M_T$, and the rearrangement $a = (1/M_T)F$ that turns the slope of a graph into a mass.

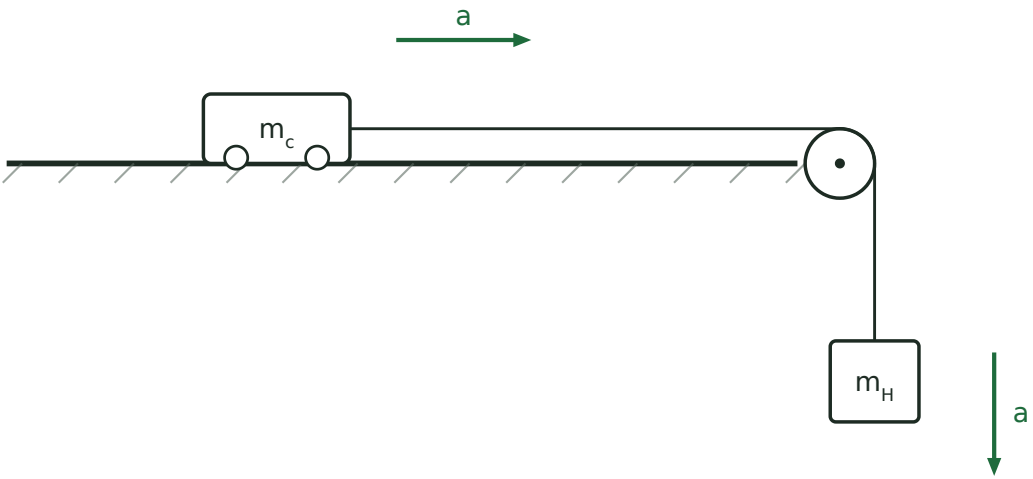
**Experimental** the setup drawing and both free-body diagrams, with the positive direction you chose for each body stated explicitly — and why total mass is held constant while the hanging mass increases.

**Analysis** the two things the lab has to confirm: that acceleration rises in proportion to force, and that the slope of that line is $1/M_T$.

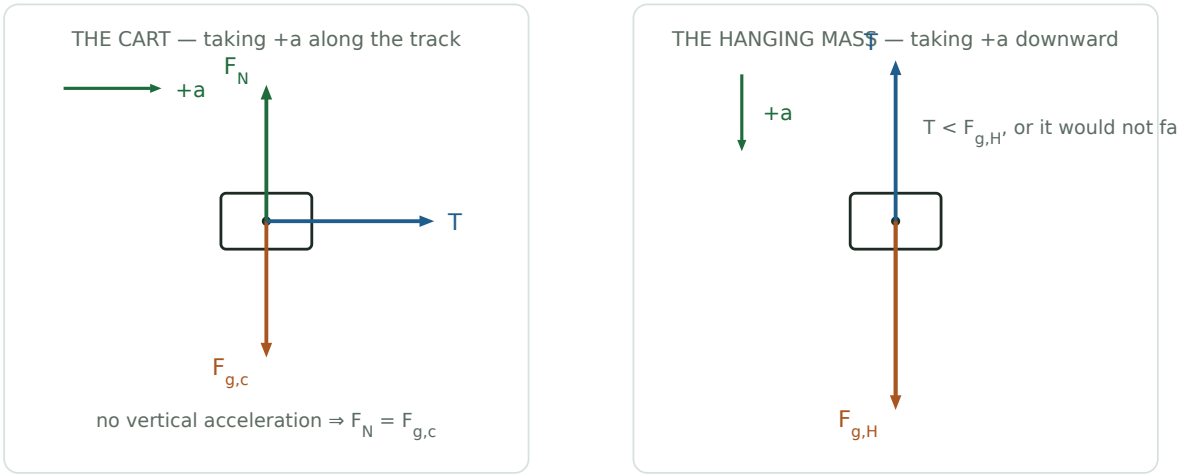
**Discussion** the three results below that your data must be checked against — the tension is less than the hanging weight, the acceleration is less than g, and one acceleration is shared by two bodies moving in different directions.

**Lab 04 — Newton's Second Law →**

## The apparatus



## Two bodies, two diagrams




1. Draw the setup in your guided notes — track, cart, pulley, string, hanging mass. Getting the geometry down is half of getting the physics right.

2. Draw both free-body diagrams, one per object, and mark +a on each.

3. Write Newton's second law for the **cart**, along the track.

4. Write Newton's second law for the **hanging mass**, taking down as positive.

## Putting them together

**Eq. 1** $T = m_c a$ — the cart — tension is its only horizontal force

**Eq. 2** $F_{g,H} - T = m_H a$ — the hanging mass — gravity wins, but only by $m_H a$

**Eq. 3** $a = \dfrac{F_{g,H}}{m_c + m_H}$

substitute the first into the second — the driving weight over the **total** mass

5. Substitute the cart equation into the hanging-mass equation and solve for a. Show every step.

## Why your prediction was probably wrong

**THE TRICK THAT MAKES THE EXPERIMENT WORK**

Do not *add* mass to the hanger. **Move** it there from the cart.

Take a small mass Δ off the cart and hang it on the string instead. The system becomes ($m_c$ − Δ) and ($m_H$ + Δ). The **total** mass in the denominator has not changed by a gram — but the driving weight in the numerator has gone up. Now the only thing varying is the net force, which is precisely the experiment Newton's second law asks for.

6. Describe the verification strategy in your own words: what do you change, what do you hold fixed, what do you measure, and **what pattern must the data show** if F = ma is telling the truth?

7. Now design the **other** half of the test — the one that checks a ∝ 1/m. What would you vary this time, and what would you keep constant?

8. Run it. Transfer the mass in equal steps, record the acceleration each time, and plot a against the hanging weight. What shape should the graph be, and what does its slope represent?

**BACK TO YOUR PREDICTION**

You predicted whether doubling the hanging mass doubles the acceleration. Was it right? Point to the term in a = $F_{g,H}$ / ($m_c$ + $m_H$) that your reasoning left out.

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# Inclined Planes — Forces on Tilted Axes

**SYNOPSIS**

A cart on a level rail stays put. Raise one end and it runs away down the ramp. Nothing about that is surprising — we have all ridden a bike up a hill and enjoyed the trip back down, and we all know gravity is responsible. What is worth asking is *how much* of gravity gets through. An incline is free fall with the gravity turned down, and the dial is the angle. That is exactly why Galileo used ramps: they slow gravity to a speed a person with a water clock can actually measure.

**BEFORE YOU READ ON — PREDICT**

A large adult and a small child each sit on a sled at the top of the same snowy hill, and both are released at the same instant.

Do they slide down at the **same rate**, or does one accelerate harder? Do they reach the bottom together, or does one arrive first?

Write down your answer and one sentence of reasoning before you read on. Galileo was the first person to settle this by measurement rather than argument.

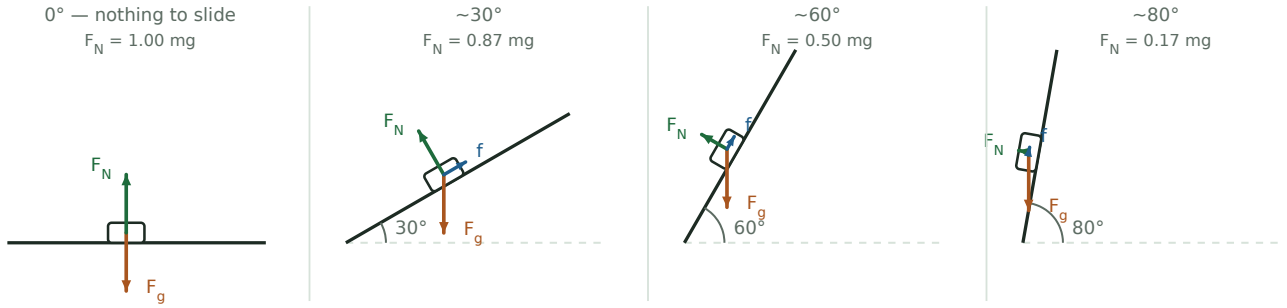
**PERFORMANCE EXPECTATIONS**

1. **Draw** the free-body diagram for an object on an incline, with the axes tilted to the slope, and say why the normal force needs no resolving while the weight does.
2. **Show** why the angle inside the force triangle equals the incline angle.
3. **Resolve** the weight into components along and perpendicular to the surface, and derive $F_N = mg \cos \theta$.
4. **Derive** $a = g \sin \theta$ for a frictionless incline, and explain why the mass does not appear in it.
5. **Check** the 0° and 90° limits against the level-surface and free-fall results.
6. **Determine** the hanging mass that holds an object in equilibrium on a frictionless incline.

7. **Derive and use** $a = g(\sin\theta - \mu\cos\theta)$ when friction acts, and interpret a negative result.

MOVEMENT ONE · THE FRICTIONLESS INCLINE

## Watch the normal force shrink



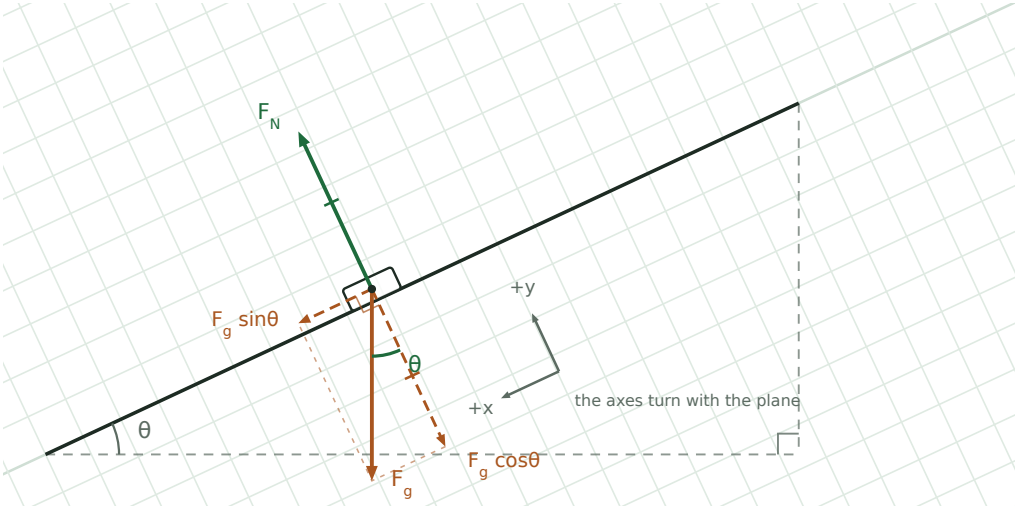

1. At **90 degrees** you are back to two forces. Why? Draw it. At that angle the surface is irrelevant to the motion — how would you describe what the object is doing?

---

## Tilt the axes, not the physics

**THE RULE THE WHOLE NOTE RESTS ON**

$$\Sigma F_x = ma_x \quad \text{and} \quad \Sigma F_y = ma_y$$




2. Which angle in the force triangle equals θ? Mark it, and **show why** it is the same as the incline angle.

3. Write the component of $F_g$ perpendicular to the surface, and the component along it, both in terms of $F_g$ and θ.

4. There is no net force along y. **Why?** Use that to write $F_N$ in terms of $F_g$ and θ.

## The key relationships

**Eq. 1** $F_N = mg\cos\theta$

the object stays on the surface, so $\Sigma F_y = 0$ — **true at any angle, with or without friction**

**Eq. 2** $F_{g,x} = mg\sin\theta$

the down-slope pull — the only part of the weight that drives the motion

**Eq. 3** $a = g\sin\theta$

**frictionless only.** Note what is missing: the mass

5. The object slides down a frictionless ramp. **Why does it accelerate at all?**

6. Use Newton's second law to show that a = g sin θ. Is this larger or smaller than the free-fall acceleration?

7. Use that equation to explain why the object **is** in free fall when the incline is vertical. How does that square with your physical intuition?

## Back to the sled

## Holding it still — the measurement

Now connect the object over a smooth pulley to a hanging mass, chosen so the object does not move.

8. Which component of $F_g$ must the hanging weight cancel for the object to be in equilibrium? Explain why it is that one and not the whole weight.

9. The object has a mass of 2 kg and the incline is at 30°. **What hanging mass holds it still?** Work it out first — then set it up and confirm your answer by measurement.

**MOVEMENT TWO · NOW PUT FRICTION BACK**

## A real ramp

**Eq. 1** $a = g(\sin\theta - \mu\cos\theta)$

sliding down a real incline — still no mass in it

**WORKED — RUBBER ON ICE, $M_K = 0.1$**

| | | |
|---|---|---|
| $\theta = 30°$ | $a = 9.8\,(0.500 - 0.1 \times 0.866)$ | $= +4.05\ m/s^2$ |
| $\theta = 5°$ | $a = 9.8\,(0.087 - 0.1 \times 0.996)$ | **$= -0.12\ m/s^2$** |

10. Set the ramp at a shallow angle with the block on it, and slowly raise one end until the block **just** breaks loose. Record that angle. What is $\mu_s$ for these two surfaces? Compare it with the value you measured by dragging in Note 5 — same block, same surfaces, two completely different methods.

**BACK TO YOUR PREDICTION**

You predicted whether the adult and the child reach the bottom together. Was it right? Point to the place in $a = g \sin\theta$ that settles it.

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# Tension Forces in a Hanging Object

**SYNOPSIS**

An object held in mid-air by two strings is the simplest two-dimensional equilibrium there is, and it hides a genuinely surprising result. The strings do not share the load in any fixed way — how hard they pull depends entirely on the *angle* they make, and as they flatten toward horizontal the tension does not level off. It runs away to infinity. This note is where you resolve vectors into components and sum them separately, which is how every mechanics problem from here on is going to work.

**BEFORE YOU READ ON — PREDICT**

A heavy sign hangs from two wires fixed to a wall. It sags in the middle and looks untidy, so you tighten both wires to pull it up closer to horizontal.

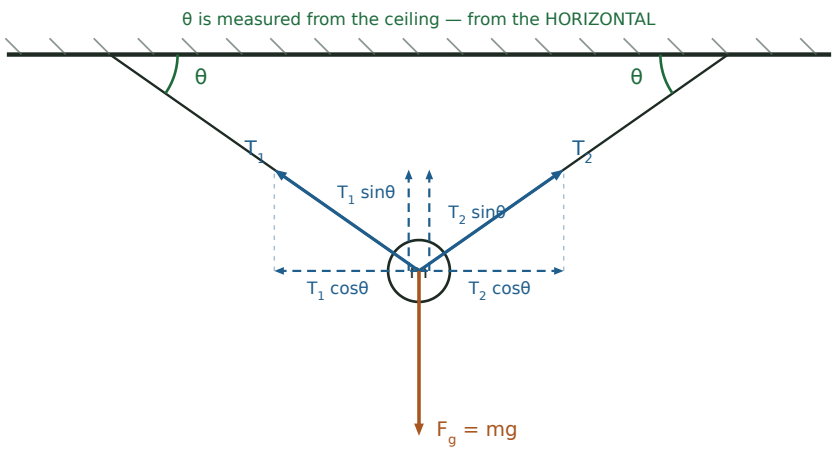
Does tightening the wires make them **safer**, or **more likely to snap**?

Commit to one, with a sentence of reasoning, before you turn the page.

**PERFORMANCE EXPECTATIONS**

1. **Draw** the free-body diagram of an object hung by two strings, and state which way a tension force acts.
2. **Resolve** each tension into horizontal and vertical components and build the component table, with the correct sign on each entry.
3. **Write** $\Sigma F_x = 0$ and $\Sigma F_y = 0$ as two independent equations, and use the first to show $T_1 = T_2$ when the angles are equal.
4. **Derive** $T = mg / (2 \sin \theta)$ and use it.
5. **Predict** how the tension changes as the strings are lengthened or shortened, and explain why a perfectly horizontal string is impossible.
6. **Explain** why the vertical components are the same at every angle while the horizontal components are not.

## The setup




1. Draw this free-body diagram yourself. Two tension forces pointing away from the object along the strings, and the weight. Mark θ at each string.

---

2. Resolve each tension into its two perpendicular components, head to tail. Which side of the right-angled triangle is adjacent to θ, and which is opposite?

---

3. Write $T_{1x}$, $T_{1y}$, $T_{2x}$ and $T_{2y}$ in terms of T and θ. **Watch the signs** — a vector pointing left takes a negative sign, and so does one pointing down.

---

## Two directions, two equations

The object is in equilibrium horizontally *and* vertically, so the forces add to zero in each direction **separately**. That is two independent equations, and it is the habit this note is really teaching. Fill the table from your own diagram.

| | $T_1$ | $T_2$ | $F_G$ | SUM |
|---|---|---|---|---|
| x-component | | | N/A | |
| y-component | | | | |

4. The strings are the same length, so the angles are the same. What does the **x-equation** then tell you about $T_1$ and $T_2$?

5. Use the **y-equation**, with $T_1 = T_2 = T$, to show that $T = mg / (2 \sin \theta)$.

## The key relationships

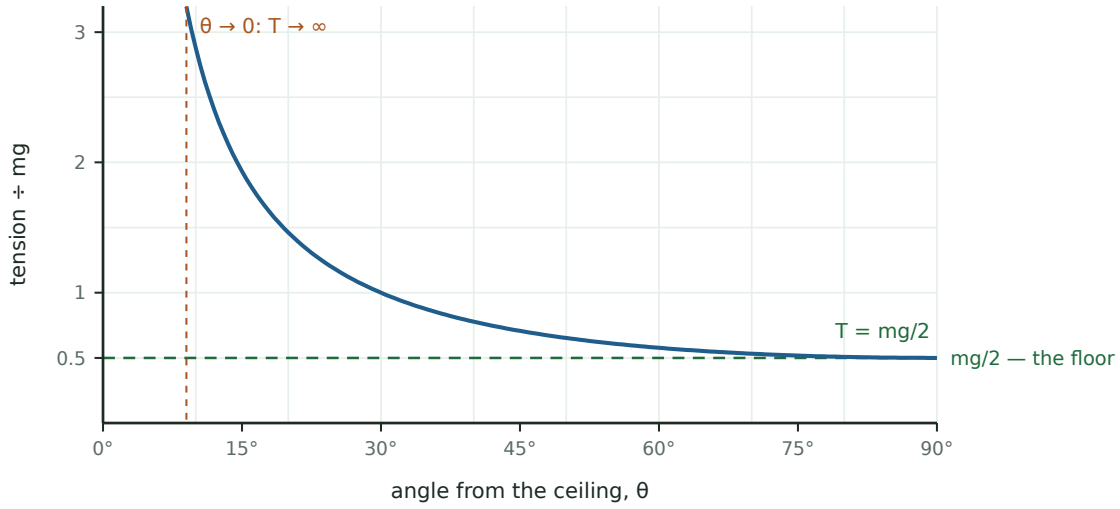
**Eq. 1** $2T\sin\theta = mg$ — the vertical components carry the whole weight, between them

**Eq. 2** $T = \frac{mg}{2\sin\theta}$ — **equal angles only.** θ from the horizontal

**Eq. 3** $T_y = \frac{mg}{2}$ — each string's vertical pull — **the same at every angle**

**Eq. 4** $T_x = \frac{mg}{2\tan\theta}$ — the horizontal pull — **unbounded** as θ → 0

## What the angle does



6. An object of mass 2 kg hangs with the strings at 30° from the ceiling. Find the tension in each string.

---

7. You replace the strings with **longer** ones, same anchor points. What happens to θ, and what happens to T? Explain using the equation.

---

8. Now use strings so short that they are almost horizontal. What happens to T, and what would that mean in practice?

---

9. If you want the tension as **small as possible**, should the strings be long or short? What is the smallest tension you could ever achieve?

---

10. You can never actually reach a 0° angle — perfectly horizontal strings. Why not? What happens to the string before you get there?

---

## Why the two components behave so differently

The *horizontal* components are a different story. They are not holding anything up — they are pulling against each other, and nothing caps how hard. As the strings flatten, each must pull harder and harder sideways just to keep its vertical share at mg/2. That is where the tension runs away, and it is why the answer to the sign question is **more likely to snap**.

11. As θ increases, what happens to the **vertical** component of the tension? And to the **horizontal** component? Justify both from the equations.

---

## Check it on the bench

12. Set up a 2 kg object hung by two strings at **30° from the horizontal**. Calculate the tension first. Then measure it, and compare. If your measured value sits below the calculation, check your angle before you doubt the physics — a few degrees matters a great deal down at this end of the curve.

---

> **WITH CALCULUS**
>
> The bench instructions above warn that “a few degrees matters a great deal down at this end of the curve.” That is a claim about a *rate*, and a rate is something you can work out rather than take on trust.

$$T(\theta) = mg / (2 \sin \theta)$$

13. Differentiate with respect to θ. Write d$T$/dθ.

14. What is the **sign** of d$T$/dθ for every angle between 0° and 90°, and what does that sign alone tell you? Check it against your answer to task 8.

15. Evaluate d$T$/dθ for the 2 kg object at θ = 60° and again at θ = 5°. How many newtons does **one degree** cost at each end? (Your derivative is per radian — convert.)

16. For small θ, replace sin θ by θ. What simple form does $T$ take? What happens to the tension if you then **halve** the angle?

17. Which factor in d$T$/dθ is responsible for the runaway — and why does the vertical component mg/2 not run away with it?

**BACK TO YOUR PREDICTION**

You predicted whether tightening the wires makes them safer or more likely to snap. Was it right? Point to the term in T = mg / (2 sin θ) that decides it.

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# Normal Force Does Not Always Equal Weight

**SYNOPSIS**

Every scale on Earth measures the *normal force* — how hard the surface pushes back on you — not your true weight. The two agree only when you are not accelerating. In this note you will alter a scale's reading with your own motion, record the normal force with a Vernier force platform while riding an elevator, and explain every feature of that graph with Newton's second law.

**BEFORE YOU READ ON — PREDICT**

You step on a bathroom scale in an elevator on the second floor and ride down to the ground floor, where the elevator comes to a stop. During that ride down, does the scale ever read *more* than your true weight?

Write down yes or no, and one sentence saying why, before you turn the page. You will check it against the recorded graph.

**PERFORMANCE EXPECTATIONS**

1. **Apply** ΣF = ma to an object on a scale, with up as positive, and solve for the normal force.
2. **State**, for a given phase of an elevator ride, whether the scale reads more than, less than, or equal to *mg*, and which way the acceleration points.
3. **Read** a normal-force-versus-time graph and label each plateau with the motion that produced it.
4. **Explain** why a scale reads zero in free fall.
5. **Determine** the normal force on an object on a level surface when an applied force is directed at an angle.

## The four experiments

1. **Spring scale:** hang a mass, then accelerate the scale up and down. Watch the reading at every phase — starting, moving, stopping.
2. **Digital scale, tilted:** the reading drops as the angle steepens. Your free-body diagram explains why — the scale reads a force component, not a mass.
3. **Bathroom scale squats:** stand still, then squat down and up. Which way does the needle deflect at each phase of the motion, and what does that tell you?
4. **The elevator ride — your data:** you ride with the Vernier force platform under your feet, starting from the second floor and descending. It traces normal force vs. time for the whole trip. Every bump and dip in that graph is Newton's second law speaking, and you will plot and tabulate it yourself below.

## The key relationships

**Eq. 1** $N = mg$

at rest *or* at constant velocity — the only case where the scale shows true weight

**Eq. 2** $N = m(g + a)$

a is signed: positive when it points up (scale reads heavy), negative when it points down (light)

**Eq. 3** $N = 0 \;\; \text{when} \;\; a = \, -g$

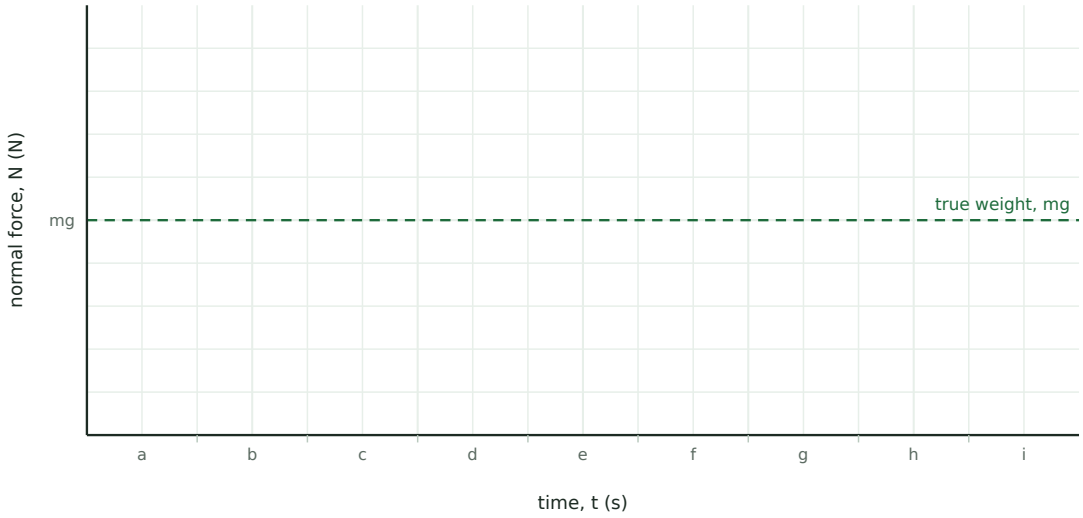
free fall — apparent weightlessness, the scale reads zero

**Eq. 4** $N = mg \pm F\sin\theta$

level surface, no vertical acceleration, applied force at angle θ: plus when you push down on it, minus when you pull up

## Record your ride

You will ride the elevator with the force platform under your feet, from the second floor down to the ground floor and back. The meter traces the normal force against time for the whole trip. Before you look at the trace, **sketch what you expect** on the axes below; then draw what the meter actually recorded on top of it, and mark where the two differ.




Normal force vs. time for one full trip. The dashed line is your true weight, *mg* — measure it standing still before you start. The letters a–i mark the nine phases of the ride listed in the table below.

## Tabulate what you recorded

Work from your own trace. Fill in the last three columns: what the scale read, which way the acceleration pointed during that phase, and its value.

| PHASE | WHAT THE ELEVATOR IS DOING | SCALE READS | DIRECTION OF A | A (M/S²) |
|---|---|---|---|---|
| a | at rest on the second floor | | | |
| b | starts moving down — speeding up | | | |
| c | moving down at constant speed | | | |
| d | still moving down, slowing to a stop | | | |
| e | at rest on the ground floor | | | |
| f | starts moving up — speeding up | | | |
| g | moving up at constant speed | | | |
| h | still moving up, slowing to a stop | | | |
| i | back at rest on the second floor | | | |

## The rule that explains the whole graph

**REMEMBER THIS ONE**

The sign of $a$ is set by whether you are **speeding up or slowing down** — not by which way you are moving.

## Free fall: the scale reads zero

**BACK TO YOUR PREDICTION**

You predicted whether the scale ever reads more than your true weight during a ride down. Was it right? Name the phase of the ride that settles it, and say which way a points there.

**WATCH · THE NOTE WORKED THROUGH**

Recording to come.  Print the guided notes (link just below), watch, and fill them in as you go.

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# Springs, Solids and Fluids: Forces You Have to Measure

**SYNOPSIS**

Nine notes of this module have done the same thing: pick an object, draw the forces on it, apply ΣF = ma. This one is different in kind. It introduces three forces — a stretched spring, a squeezed solid, the air pushing back on a falling body — whose strengths **do not follow from Newton's laws at all**. Each carries a constant that nobody can derive and everybody has to measure. That is not a gap in the theory. It is how physics meets real materials.

**BEFORE YOU READ ON — PREDICT**

A skydiver steps out of an aircraft. From **N4** you know that near the Earth every falling object accelerates at 9.8 m/s², whatever its mass.

Does the skydiver still accelerate at 9.8 m/s² ten seconds later? Thirty?

Answer, with one sentence saying what would have to be true for your answer to be right, before you read on.

**PERFORMANCE EXPECTATIONS**

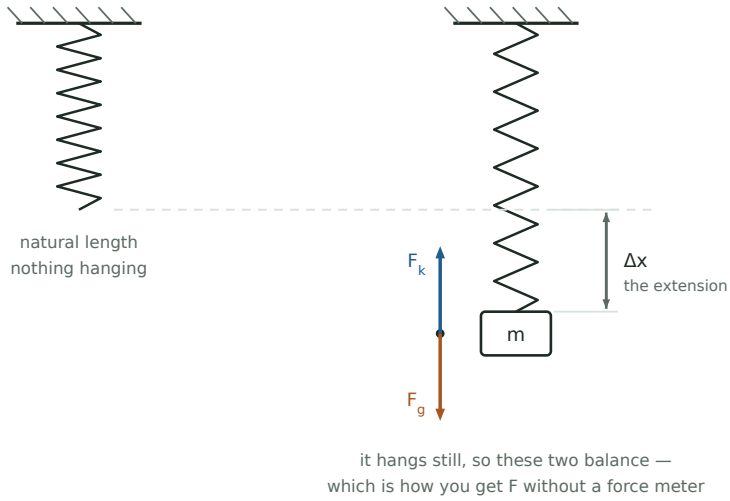
1. **Relate** the force in a stretched or compressed spring to its extension, and **distinguish** the extension from the spring's total length.
2. **Determine** k from the slope of a force-against-extension graph, state its unit, and **interpret** it as stiffness.
3. **Apply** the stress-strain relations for compression, stretch and shear, and **say** what Y and S describe.
4. **Explain** how terminal velocity arises from a force that grows with speed competing with one that does not.
5. **Compute** a terminal velocity from mass, area, shape and fluid density.
6. **Explain** why k, Y, S and $C_D$ have to be measured rather than derived, and **name** the constant from earlier in this module that is the same kind of quantity.

MOVEMENT 1 — A DIFFERENT KIND OF FORCE LAW

# Laws you derive, and laws you measure

MOVEMENT 2 — SPRINGS

## Hooke’s law



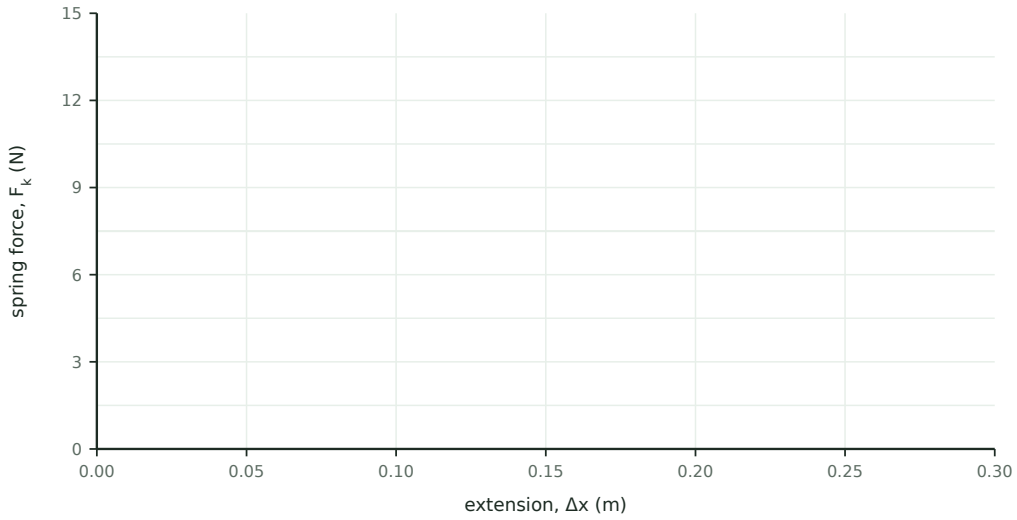

**Eq. 1** $F_k = k\Delta x$

double the extension and the spring pulls back twice as hard

## Measuring k

Hang known weights, record the extension each time, and plot one against the other. If Hooke’s law is right the points fall on a straight line through the origin, and the slope of that line is the spring constant.

| HANGING WEIGHT, F (N) | EXTENSION, ΔX (CM) | ΔX IN METRES |
|---|---|---|
| 2 | | |
| 4 | | |
| 12 | | |




Plot force against extension and draw the best straight line. Note the units on the horizontal axis — convert before you plot, not after.

1. Find the slope from your line, and give its unit.

2. You have two springs. One has k = 50 N/m and the other k = 500 N/m. Which is harder to stretch, and by what factor? Sketch both on the same axes.

3. A spring with a larger k gives a steeper line. Say in one sentence what “steeper” means physically here.

**WITH CALCULUS**

You found $k$ as the slope of a line you drew by eye. Written exactly, that slope is a derivative:

$$k = \mathrm{d}F_\mathrm{k} / \mathrm{d}(\Delta x)$$

Which is a stronger statement than it looks, because a derivative is defined at a *point* and your line was drawn across a whole range.

1. If the $F$–$\Delta x$ graph really is a straight line, what is true about $\mathrm{d}F_\mathrm{k}/\mathrm{d}(\Delta x)$ at every point on it? That statement is what “$k$ is a constant” actually means.

2. Sketch the graph for a spring pulled well past its elastic limit. Mark the stretch where $\mathrm{d}F_\mathrm{k}/\mathrm{d}(\Delta x)$ is constant, and say what is happening to that derivative beyond it.

3. So state the elastic limit as a condition on the **derivative** rather than on the spring.

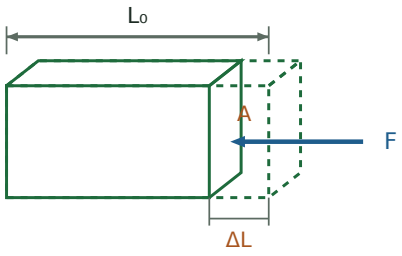
4. $Y$ and $S$ were defined the same way, as stress over strain. Write each as a derivative. What plays the part of $\Delta x$ in each one?

5. A constant obtained this way — the slope of a curve across the stretch where the curve is near enough straight — is called a **local linearization**. Using your sketch from task 2, say what the word *local* is doing in that phrase.

MOVEMENT 3 — SOLIDS

## Hooke’s law, spread over an area

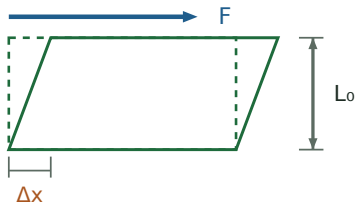
COMPRESSION AND STRETCH




A is the face the force pushes on

$F / A = Y (\Delta L / L_0)$

SHEAR




the top face slides; the block leans

$F / A = S (\Delta x / L_0)$

**IN BOTH CASES**

stress = modulus × strain

Y is the **Young’s modulus** and S the **shear modulus**. Both are properties of the material, not of the particular block — steel has the same Y whatever shape you cut it into, which is exactly why writing the law in terms of F/A and $\Delta L/L_0$ rather than F and ΔL was worth the trouble.

1. Compare $F = k\Delta x$ with $F/A = Y(\Delta L/L_0)$. Which symbol in the second plays the part that k plays in the first? What did the other two changes buy you?

2. Two bars of the same steel, same length, one twice the cross-sectional area. Under the same force, which shortens more, and by what factor?

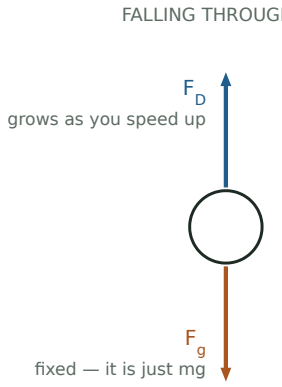
3. A material with a large Y — is it stiff or floppy? Give an example of each.

MOVEMENT 4 — FLUIDS

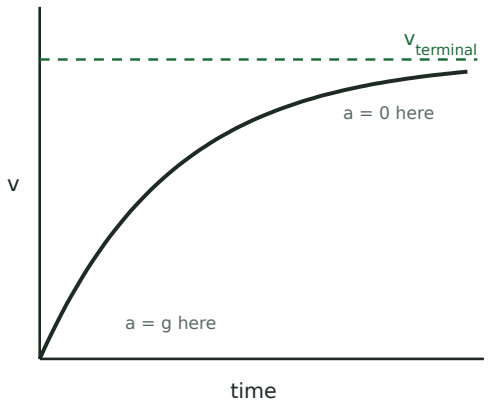
## Drag, and why falling stops speeding up

**Eq. 1** $F_D = \frac{1}{2} C_D \rho A v^2$

$C_D$ shape · ρ fluid density · A frontal area




the only force that changes is the drag




speed stops rising when the two forces match

4. Write ΣF = ma for the falling body, taking down as positive, with both forces in it.

5. Set the acceleration to zero and solve for v. This is the terminal velocity.

**Eq. 1** $v_{\text{term}} = \sqrt{\dfrac{2mg}{C_D \rho A}}$

the speed at which drag has grown to equal the weight

### TASK 6 — A SKYDIVER

1. Take m = 100 kg, frontal area A = 2 m², air density ρ = 1.3 kg/m³, and $C_D$ = 0.7 for a spread-eagled body. Find the terminal velocity in m/s.

2. Convert it to km/h.

3. Skydivers report roughly 200 km/h. Yours will not match. Is the disagreement small enough to call this model useful? Name **one** number in the calculation you would want to know better.

4. The same skydiver goes head-down, cutting A roughly in half. What happens to the terminal velocity — and by what factor, exactly?

### WITH CALCULUS

Everything so far has been about the *end* of the fall. The hook asked about the middle of it — whether the skydiver is still gaining speed after ten seconds, and after thirty. Answering that means writing the second law as an equation in d*v*/d*t* and solving it.

1. Take **down as positive**. Write Newton's second law for the falling skydiver with drag included, as an equation containing d*v*/d*t*.

2. Set d*v*/d*t* = 0 and solve for *v*. Compare with the $v_{\text{term}}$ you used in Task 6. What have you just shown about where that formula came from?

3. **Separate the variables and integrate**, taking $v = 0$ at $t = 0$, to show that $v(t) = v_{\text{term}} \tanh(gt / v_{\text{term}})$. Write your $v$-integral in terms of $v_{\text{term}}$ first — the standard form is easier to recognise that way.

4. Check your solution at the two ends: what is $v$ at $t = 0$, and what does $v$ approach as $t$ grows?

5. $v_{\text{term}}/g$ has units of time. Work it out from your own Task 6 answer. Roughly how many of those does the fall need to reach 99% of terminal velocity?

6. Sketch $v$ against $t$, marking $v_{\text{term}}$ and your timescale on the axes. Now answer the question in the hook from the graph rather than in words: still accelerating at ten seconds? At thirty?

**BACK TO YOUR PREDICTION**

You predicted whether the skydiver is still accelerating at 9.8 m/s² after ten seconds and after thirty. Was it right? Point to the moment on the speed-against-time graph where your answer changes, and name the force that changed it.

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